

CS 237B: Principles of Robot Autonomy II
Problem Set 4: System Verification
Due 11:59PM March 11th, 2022

You will submit your homework to Gradescope. Your submission will consist of a single pdf with your answers for written questions (denoted by the ✍ symbol).

Important: Remember, your written part must be typeset (e.g., L^AT_EX or Word).

Problem 1

- (i)  Read the paper by Eppner et al [1] on lessons learned from the Amazon Picking Challenge. Review the paper by providing
- a summary of the main points in your own words,
 - a discussion of the advantages and disadvantages of the overall approach from the viewpoint of designing system architectures

Problem 2: Specifications and Model Checking

(i)  Explain if the linear temporal logic specification $\mathbf{F} (\psi_1 \Rightarrow \psi_2)$ is equivalent to $\mathbf{true} \mathbf{U} (\psi_1 \vee \neg\psi_2)$.

(ii)  For each specification below, explain if it formalizes the following statement or not: "One might enter Stanford Stadium in the next state only if they have a game ticket or they are a Stanford student in the current state."

Note: The predicate "GameTicket" indicates having a game ticket, "StanfordStudent" indicates being a Stanford student, and "StanfordStadium" indicates entering the Stadium (not eligibility to enter the stadium!).

- (a) $\mathbf{G} ((\text{GameTicket} \vee \text{StanfordStudent}) \mathbf{U} \text{StanfordStadium})$
- (b) $\mathbf{G} ((\text{GameTicket} \vee \text{StanfordStudent}) \Rightarrow \mathbf{X} \text{StanfordStadium})$
- (c) $\mathbf{G} ((\neg \mathbf{X} \text{StanfordStadium}) \Rightarrow \neg(\text{GameTicket} \vee \text{StanfordStudent}))$
- (d) $\mathbf{G} ((\neg \text{GameTicket} \wedge \neg \text{StanfordStudent}) \Rightarrow \mathbf{X} \neg \text{StanfordStadium})$
- (e) $\mathbf{G} (\text{StanfordStadium} \Rightarrow (\text{GameTicket} \vee \text{StanfordStudent}))$
- (f) $\mathbf{G} ((\mathbf{X} \text{StanfordStadium}) \Rightarrow (\text{GameTicket} \vee \text{StanfordStudent}))$

(iii)  For each specification below, explain if it formalizes the following statement: "One can take CS237B at most once."

Note: The predicate "CS237B" indicates taking CS237B (not eligibility to take it!).

- (a) $(\mathbf{F} \text{CS237B}) \mathbf{U} (\mathbf{G} \neg \text{CS237B})$
- (b) $\mathbf{G} ((\mathbf{F} \text{CS237B}) \Rightarrow \neg \text{CS237B})$
- (c) $\mathbf{G} ((\mathbf{X} \mathbf{F} \text{CS237B}) \Rightarrow \neg \text{CS237B})$
- (d) $\text{CS237B} \wedge \mathbf{X} \mathbf{G} \neg \text{CS237B}$
- (e) $\text{CS237B} \mathbf{U} (\mathbf{G} \neg \text{CS237B})$
- (f) $(\mathbf{F} \text{CS237B}) \wedge (\mathbf{F} \mathbf{G} \neg \text{CS237B})$

(iv)  Consider a state machine M whose transitions are:

$$s_t = \begin{cases} 0 & \text{if } a_t \leq b_t, \\ 1 & \text{otherwise.} \end{cases}$$

$$a_{t+1} = \begin{cases} a_t + 1 \pmod{10} & \text{if } s_t = 0, \\ a_t & \text{otherwise.} \end{cases}$$

$$b_{t+1} = \begin{cases} b_t + 1 \pmod{10} & \text{if } s_t = 1, \\ b_t & \text{otherwise.} \end{cases}$$

for time index $t \geq 0$ with $a_0 = b_0 = 0$. Explain if the following specifications are true or not:

- (a) $\mathbf{FG} (s = 0)$
- (b) $\mathbf{GF} (s = 1)$

References

- [1] C. Eppner, S. Höfer, R. Jonschkowski, R. Martín-Martín, A. Sieverling, V. Wall, and O. Brock, “Four aspects of building robotic systems: lessons from the amazon picking challenge 2015,” *Autonomous Robots*, vol. 42, p. 1459–1475, 2018. [Online]. Available: <https://link-springer-com.stanford.idm.oclc.org/article/10.1007/s10514-018-9761-2>